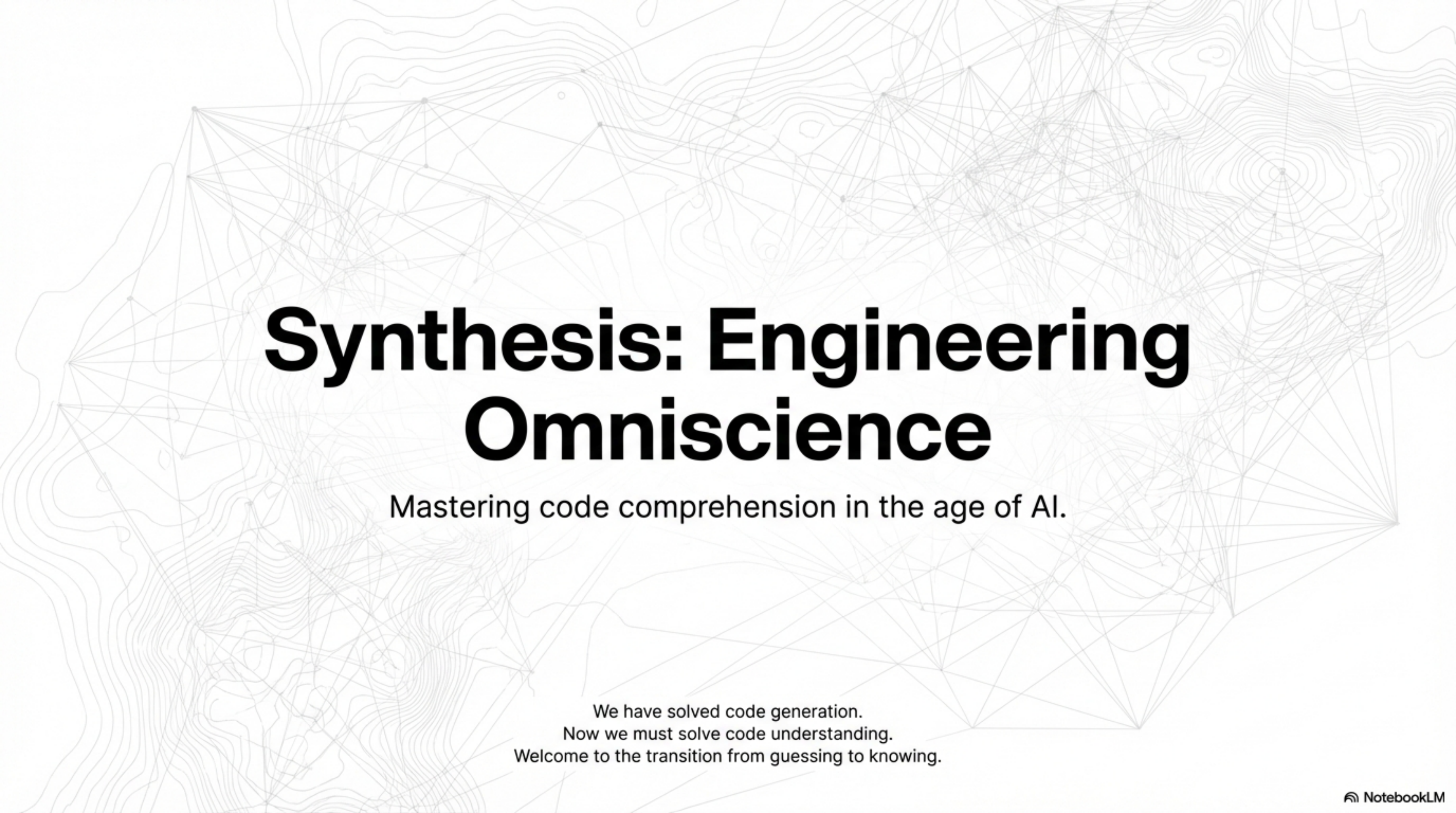


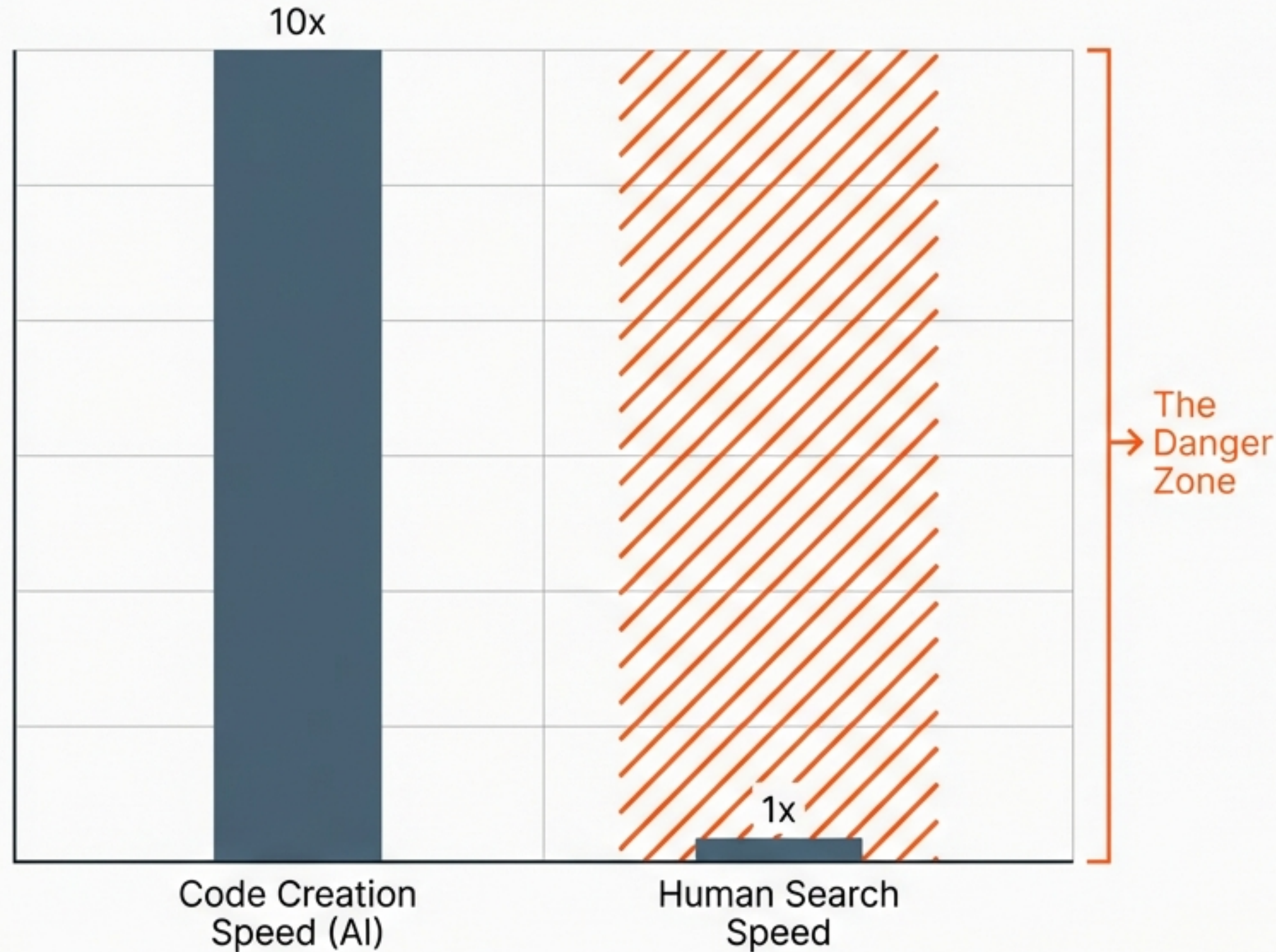


Synthesis: Engineering Omniscience

Mastering code comprehension in the age of AI.

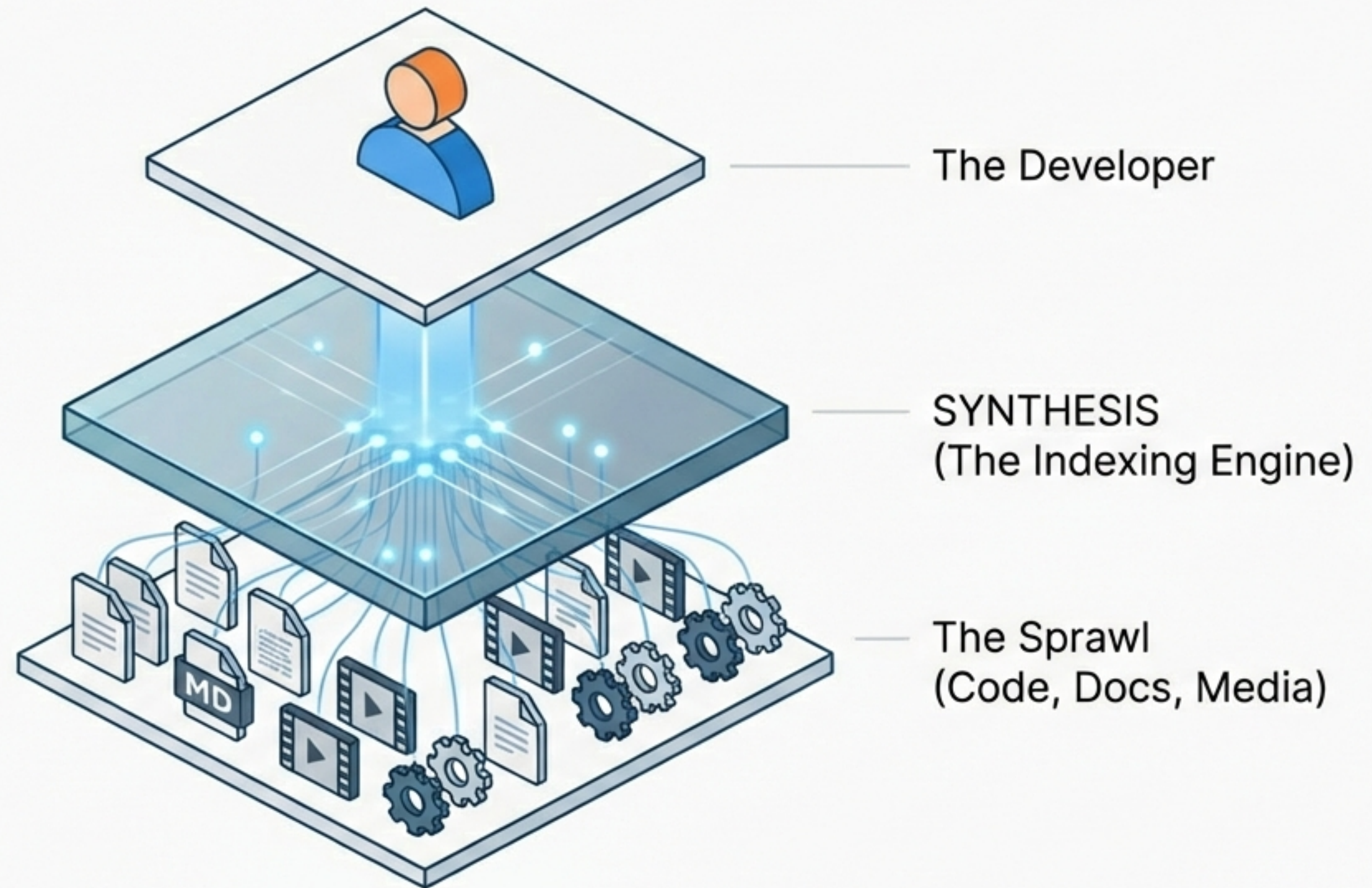
We have solved code generation.
Now we must solve code understanding.
Welcome to the transition from guessing to knowing.

The Comprehension Bottleneck



- 40-60% of developer time is spent searching, not building.
- Your IDE sees one project. You work across ten.
- The result: Wasted AI investment and velocity death.

Synthesis is Knowledge Infrastructure



It's not just grep. It's deep context, available in seconds.

Five Minutes to Indexed Reality

Prerequisites:

- Java 17+ Required

Installation:

```
run: curl -fsSL .../install.sh | bash
```

Verification:

```
run: synthesis --version
```

```
user@dev:~/project $ synthesis init
```

✓ Workspace initialized: /home/you/your-project
i Config: .synthesis/config.yaml

```
user@dev:~/project $ synthesis scan
```

Scanning...

100%

```
✓ Indexed 1,500 files in 8s (200-300 files/sec).
```

Search Beyond Grep

🔍 synthesis search "authentication"



AuthService.java - src/main/java/com/app/auth
public class AuthService implements...



auth-flow.md - docs/architecture
The authentication flow uses JWT tokens ...



security.yaml - config/prod
auth_timeout: 3600

Unified Search: Code, Docs, and Configs simultaneously.

Relevance Ranking: Context over string matching.

Exercise 01: The Discovery Hunt

Execute these commands on your codebase.

01

Find a Feature

```
synthesis search '[feature name]'
```

02

Find the Debt

```
synthesis search 'TODO'
```

03

Find the Decay

```
synthesis search '@Deprecated'
```

04

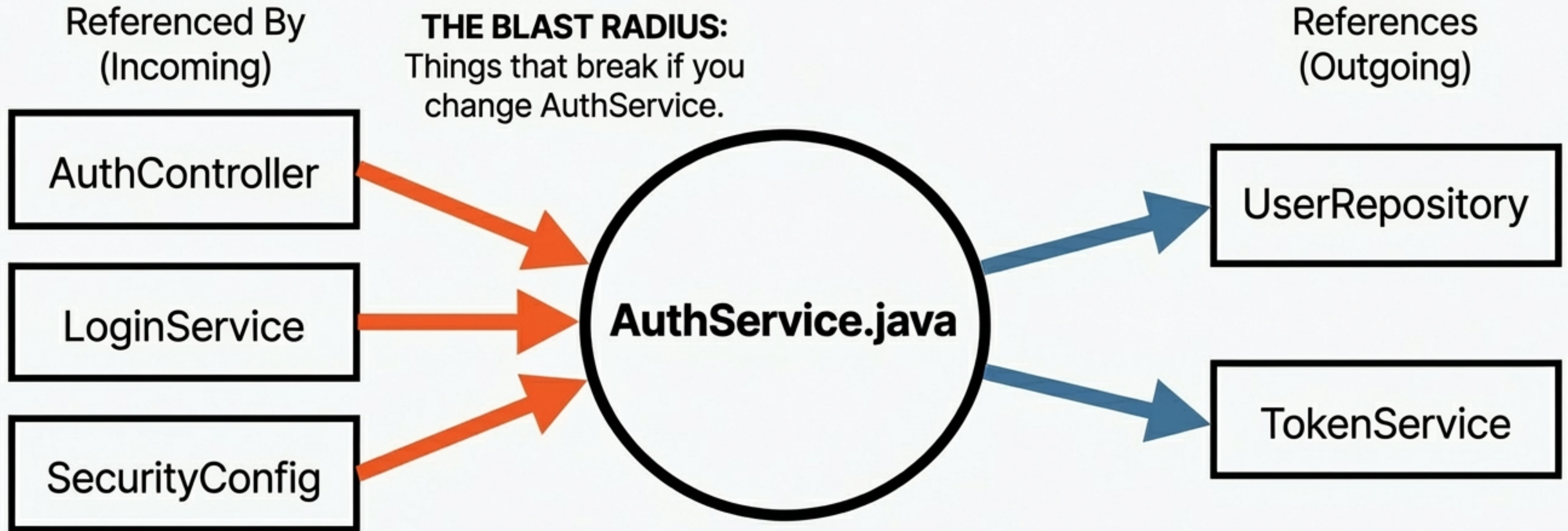
Find the Config

```
synthesis search 'ENV_VAR'
```

Insight: What surprised you? Most teams find dead code in the first 60 seconds.

Mapping the Blast Radius

synthesis relate 'AuthService.java'



Exercise 02: Predicting the Future

SCENARIO: You are about to refactor a critical file.

ACTION: Run ``synthesis relate 'YourCriticalFile.java'``



High Risk. 20+
Incoming References.
Tread carefully.

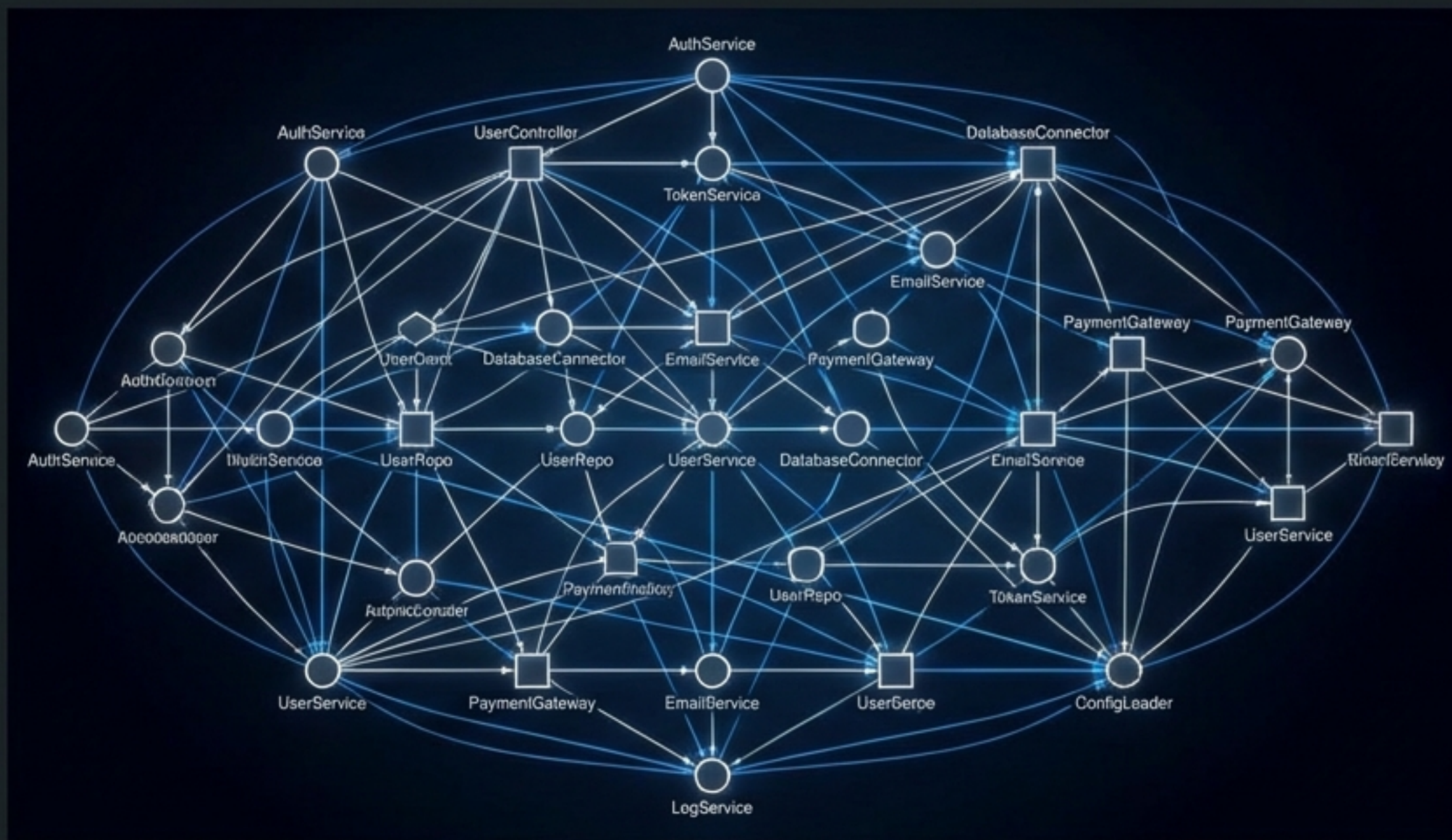


Dead Code.
0 Incoming References.
Safe to delete?

The Lesson: Never refactor without checking the blast radius first.

The Architecture is the Code

```
synthesis graph --modules --format mermaid > architecture.md
```

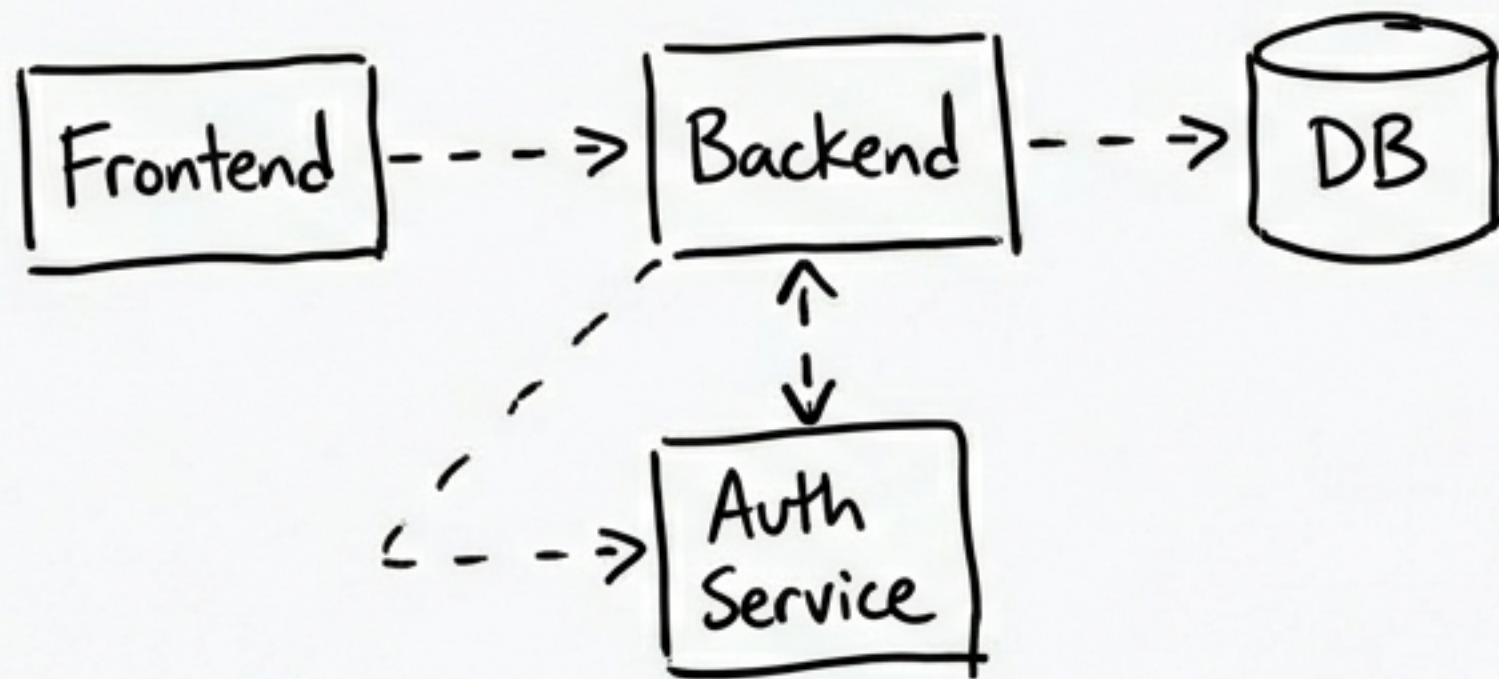


- Stop relying on stale Confluence pages.
- Stop drawing whiteboard diagrams that expire in a week.
- Let the code draw itself.

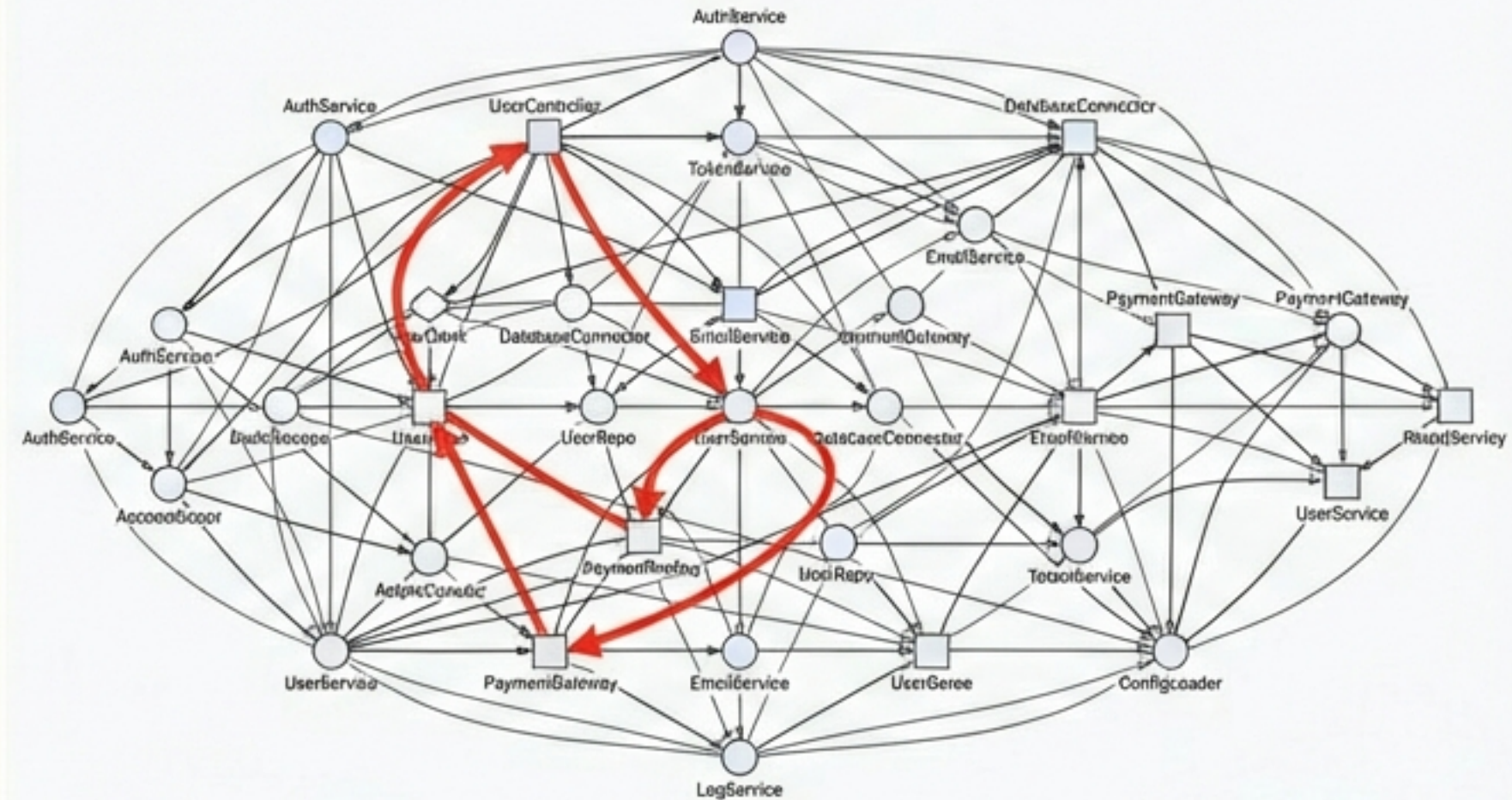
Formats: Mermaid,
PNG, SVG, DOT.

Exercise 03: Truth Serum

What you think the architecture is



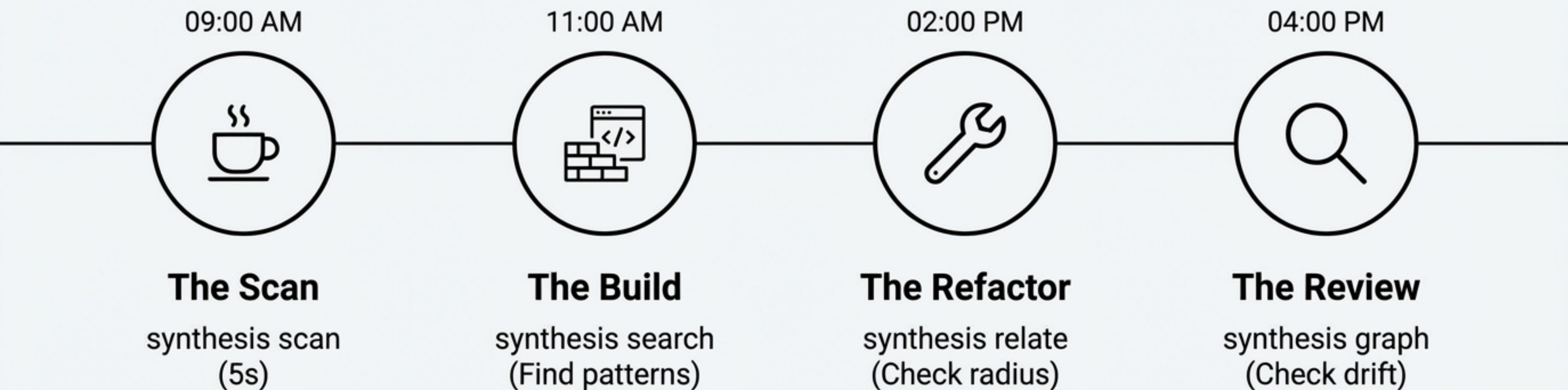
What Synthesis sees



1. Generate graph.
2. View in mermaid.live.
3. Spot the circular dependencies.

The Synthesis Loop

Search before you Build.



Scaling to the Team

Onboarding



Create 'New Hire Guides' by searching for key terms and generating dependency maps of core services.

Tech Debt Audits



Automated reporting on 'TODO', 'FIXME', and '@Deprecated' tags across the entire repo.

CI/CD Integration



Safety checks in Pull Requests. Require 'relate' output for changes to shared services.

Pro-Tip: Alias: alias syn='synthesis'

Troubleshooting & Optimization

Issue	Fix
> Error: Not a workspace	Run <code>`synthesis init`</code> in the root folder.
> Scan is slow (>1 min)	Exclude <code>'node_modules/'</code> or <code>'target/'</code> in <code>.synthesis/config.yaml</code>
> No Results found	Check your <code>'includePatterns'</code> in config.
> Java Error	Ensure <code>`java -version`</code> is 17+.

Optimization: Use `'Watch Mode'` for continuous indexing.

Making it Stick

Today

Install and run the first scan.

Tomorrow

Set up a Git Hook to auto-scan on commit.

Next Week

Establish Team Norms: Attach graphs to major PRs.

Resources

- Quick Start Guide
- Full User Documentation
- DevOps Guide for CI/CD

Stop Searching. Start Synthesizing.

```
synthesis init           // Initialize  
synthesis scan          // Update Index  
synthesis search 'query' // Find Knowledge  
synthesis relate 'file'  // Map Dependencies  
synthesis graph --modules // Visualize Arch
```

Engineering Omniscience is now installed. Good luck.